

Ali Mohamed-Ali | Mechanical Engineer

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Education

North Carolina State University – BS in Mechanical Engineering	May 2025
Davidson-Davie Community College – Certificate of Electrical & Mechatronics Engineering	May 2021

Professional Experience

Product Design Engineer , Trion IAQ/ADTI – Sanford, NC	June 2025 – April 2026
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- Led New Product Introduction (NPI) lifecycles for top-selling residential and commercial product lines, utilizing SolidWorks to design cost-effective components optimized for state-of-the-art sheet metal fabrication
- Executed comparative air-leakage lab tests on current generation versus competitor duct-mount cabinets to inform R&D design targets focused on manufacturing efficiency, thermal performance, and customer needs
- Developed proprietary filter adapter solutions for high-value commercial initiatives (millions in pipeline scope), bridging internal fan-filter units with customer-specific dimensions through rigorous cross-functional review
- Reverse-engineered precision ceiling-mount adapters for industrial-grade fan-filter units, ensuring dimensional compatibility with existing internal systems while maintaining structural integrity and load capacity
- Facilitated cross-functional design reviews between Product Engineering, Manufacturing Operations, and Sales leadership to integrate real-time shop-floor feedback into CAD iterations during the development phase
- Optimized manufacturing costs and efficiency by tailoring cabinet wrapper designs for specific state-of-the-art sheet metal bending machines, reducing waste and improving production throughput
- Managed technical documentation and project milestones, maintaining detailed Gantt charts, and formal reports to track fabrication timelines, material procurement, and budget adherence during R&D

Humanoid Robotics Research Assistant , HIER Lab, NC State – Raleigh, NC	October 2024 – May 2025
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- Researched joint designs to improve torque-to-energy ratio and range of motion in humanoid legs
- Modeled and prototyped linkage-based mechanisms in SolidWorks, 3D printing iterations for testing
- Improved prototype by adding an additional degree of freedom to the foot, enhancing rotational mobility
- Gained hands-on experience with basic Arduino electronics and circuitry for integrating sensors and motors
- Documented work in formal technical reports detailing design evolution and outcomes
- Hosted two seminars to present topics in robotic locomotion, including linkage-driven actuation and dynamic balance

Mechanical Design Engineering Intern , iFab Corp. – Gastonia, NC	May 2024 – Aug 2024
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- Designed low-cost welding fume extractor to reduce health risks for fabrication workers
- Interviewed operators to gather feedback and align ergonomic improvements with user needs
- Consulted with EHS and engineering leadership to refine CAD prototypes and BOM through multiple iterations
- Achieved ~58% cost reduction by developing in-house alternative to \$6,000 commercial solutions
- Redesigned grinding stations to improve airflow and usability with SolidWorks and GD&T

Academic/Extracurricular Experience

Automatic Low-Profile Car Jack for Formula SAE , Senior Design, NC State – Raleigh, NC	August 2024 – May 2025
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- Collaborated with a 6-person team to design a remote-controlled scissor-lift jack for rapid pit-lane lifting of low-clearance vehicles
- Modeled the full assembly in SolidWorks and validated structural integrity using FEA simulations
- Fabricated, assembled, and tested the prototype using MIG welding, mechanical hardware, and winch-driven actuation
- Created a project Gantt chart and managed budgeting, fabrication timeline, and material procurement
- Maintained a detailed engineering journal and presented project milestones through technical reports/formal presentations

Skills

Design & Simulation: SolidWorks, ANSYS, MATLAB

Technical & Fabrication: GD&T, 3D Printing (FDM), MIG Welding, Microsoft Office